

EduSaaS PDDD - Part 2

Chapters 6 to 11: Domain Table Dictionary

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Schema: education | Database: PostgreSQL | Architecture: Backend-owned persistence, stateless AI services

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Chapter 6: Learning Domain

This chapter documents all Learning tables, columns, constraints and purposes.

Table: education.courses

Column	Type	Constraint	Description
id	UUID	PK	Course identifier
title	VARCHAR(255)	NOT NULL	Course title
description	TEXT	NULL	Description
provider	VARCHAR(255)	NULL	Provider
category	VARCHAR(255)	NULL	Category
difficulty	VARCHAR(50)	NULL	Difficulty
status	VARCHAR(50)	DEFAULT active	Course status
educator_id	UUID	FK NULL	References users.id
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.lessons

Column	Type	Constraint	Description
id	UUID	PK	Lesson id
course_id	UUID	FK	References courses.id
title	VARCHAR(255)	NOT NULL	Lesson title
video_url	TEXT	NULL	Video URL
duration	INTEGER	NULL	Duration
order_index	INTEGER	DEFAULT 0	Sequence
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.quizzes

Column	Type	Constraint	Description
id	UUID	PK	Quiz id
lesson_id	UUID	FK	References lessons.id
questions	JSONB	NOT NULL	Quiz questions payload
passing_score	INTEGER	DEFAULT 60	Passing score

Table: education.assignments

Column	Type	Constraint	Description
id	UUID	PK	Assignment id
lesson_id	UUID	FK	References lessons.id
instructions	TEXT	NULL	Instructions
submission_link	TEXT	NULL	Submission URL
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.enrollments

Column	Type	Constraint	Description
id	UUID	PK	Enrollment id
user_id	UUID	FK	References users.id
course_id	UUID	FK	References courses.id
status	VARCHAR(50)	DEFAULT active	Enrollment status
completion_percentage	DOUBLE PRECISION	DEFAULT 0	Completion %
enrolled_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Enrollment date

Table: education.progress

Column	Type	Constraint	Description
id	UUID	PK	Progress id

user_id	UUID	FK	References users.id
lesson_id	UUID	FK	References lessons.id
watched_duration	INTEGER	DEFAULT 0	Watched duration
quiz_score	INTEGER	NULL	Quiz score
assignment_status	VARCHAR(50)	DEFAULT pending	Assignment status
completion_flag	BOOLEAN	DEFAULT false	Completion flag
updated_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Updated
completed_at	TIMESTAMP(6)	NULL	Completed

Table: education.certificates

Column	Type	Constraint	Description
id	UUID	PK	Certificate id
user_id	UUID	FK	References users.id
course_id	UUID	FK	References courses.id
certificate_code	VARCHAR(255)	UNIQUE	Certificate code
issued_date	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Issued

Table: education.achievements

Column	Type	Constraint	Description
id	UUID	PK	Achievement id
user_id	UUID	FK	References users.id
badge_name	VARCHAR(255)	NOT NULL	Badge name
milestone	TEXT	NULL	Milestone
certificate_id	UUID	FK NULL	References certificates.id
earned_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Earned

Table: education.tasks

Column	Type	Constraint	Description
id	UUID	PK	Task id
user_id	UUID	FK	References users.id
course_id	UUID	FK NULL	References courses.id
title	VARCHAR(255)	NOT NULL	Task title
due_date	TIMESTAMP(6)	NULL	Due date
status	VARCHAR(50)	DEFAULT pending	Task status
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
completed_at	TIMESTAMP(3)	NULL	Completed

Table: education.recommendations

Column	Type	Constraint	Description
id	UUID	PK	Recommendation id
user_id	UUID	FK	References users.id
course_id	UUID	FK	References courses.id
reason	TEXT	NULL	Recommendation reason
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
predicted_rating	DOUBLE PRECISION	AI ADD	Predicted preference
similarity_score	DOUBLE PRECISION	AI ADD	Similarity score
confidence_score	DOUBLE PRECISION	AI ADD	Confidence score
prerequisite_completed	BOOLEAN	AI ADD	Prerequisite flag
rank	INTEGER	AI ADD	Rank
updated_at	TIMESTAMP	AI ADD	Updated

Chapter 7: Assessment & Skills Domain

This chapter documents all Assessment & Skills tables, columns, constraints and purposes.

Table: education.domain_roles

Column	Type	Constraint	Description
domain_role_id	UUID	PK	Domain role id
domain_name	VARCHAR(100)	UNIQUE	Domain name
category	VARCHAR(50)	NULL	Category
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.skills

Column	Type	Constraint	Description
skill_id	SERIAL/INTEGER	PK	Skill id
skill_name	VARCHAR(100)	UNIQUE	Skill name
category	VARCHAR(50)	NULL	Category
description	TEXT	NULL	Description
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.domain_required_skills

Column	Type	Constraint	Description
domain_required_skill_id	SERIAL	PK	Mapping id
domain_role_id	UUID	FK	References domain_roles
skill_id	INTEGER	FK	References skills
required_level	SMALLINT	NOT NULL	Required level

Table: education.assessments

Column	Type	Constraint	Description
id	UUID	PK	Assessment id
user_id	UUID	FK	References users
type	VARCHAR(100)	NOT NULL	Assessment type
score	DOUBLE PRECISION	NOT NULL	Score
answers	JSONB	NULL	Answers
date_taken	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Taken
career_goal	TEXT	NULL	Career goal
is_initial	BOOLEAN	DEFAULT false	Initial flag
total_marks	INTEGER	NULL	Total
obtained_marks	INTEGER	NULL	Obtained
completed	BOOLEAN	DEFAULT false	Completed
started_at	TIMESTAMP(6)	NULL	Started
completed_at	TIMESTAMP(6)	NULL	Completed at

Table: education.gap_reports

Column	Type	Constraint	Description
id	UUID	PK	Report id
user_id	UUID	FK	References users
readiness_score	DOUBLE PRECISION	NOT NULL	Readiness
missing_skills	TEXT[]	NULL	Missing skills
recommendations	JSONB	NULL	Recommendations
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Updated

Table: education.difficulty_levels

Column	Type	Constraint	Description
difficulty_id	SERIAL	PK	Difficulty id
difficulty_name	VARCHAR(20)	UNIQUE	Difficulty
difficulty_order	INTEGER	UNIQUE	Order

Table: education.questions

Column	Type	Constraint	Description
question_id	SERIAL	PK	Question id
skill_id	INTEGER	FK	References skills
difficulty_id	INTEGER	FK	References difficulty_levels
question_text	TEXT	NOT NULL	Question
option_a	TEXT	NOT NULL	Option A
option_b	TEXT	NOT NULL	Option B
option_c	TEXT	NOT NULL	Option C
option_d	TEXT	NOT NULL	Option D
correct_option	CHAR(1)	NOT NULL	Correct option
marks	INTEGER	DEFAULT 1	Marks
is_active	BOOLEAN	DEFAULT true	Active
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.quiz_sessions

Column	Type	Constraint	Description
session_id	SERIAL	PK	Session id
user_id	UUID	FK	References users
domain_role_id	UUID	FK	References domain_roles
start_time	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Start
end_time	TIMESTAMP(6)	NULL	End
status	VARCHAR(20)	DEFAULT In Progress	Status
total_questions	INTEGER	DEFAULT 50	Total
questions_answered	INTEGER	DEFAULT 0	Answered
current_skill_id	INTEGER	NULL	Current skill

Table: education.student_answers

Column	Type	Constraint	Description
answer_id	SERIAL	PK	Answer id
session_id	INTEGER	FK	References quiz_sessions
skill_id	INTEGER	FK	References skills
question_id	INTEGER	FK	References questions
difficulty_id	INTEGER	FK	References difficulty_levels
selected_option	CHAR(1)	NOT NULL	Selected
correct_option	CHAR(1)	NOT NULL	Correct
is_correct	BOOLEAN	NOT NULL	Correct flag
marks_awarded	INTEGER	DEFAULT 0	Marks
answered_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Answered

Table: education.student_skill_results

Column	Type	Constraint	Description
result_id	SERIAL	PK	Result id
session_id	INTEGER	FK	References quiz_sessions
skill_id	INTEGER	FK	References skills
obtained_score	INTEGER	NOT NULL	Obtained
maximum_score	INTEGER	NOT NULL	Maximum
percentage	DECIMAL(5,2)	NOT NULL	Percentage
skill_level	SMALLINT	NOT NULL	Level

created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
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Table: education.quiz_state

Column	Type	Constraint	Description
session_id	INTEGER	PK/FK	References quiz_sessions
skill_id	INTEGER	PK/FK	References skills
current_difficulty	INTEGER	DEFAULT 1	Difficulty
correct_streak	INTEGER	DEFAULT 0	Correct streak
wrong_streak	INTEGER	DEFAULT 0	Wrong streak
questions_answered	INTEGER	DEFAULT 0	Answered
obtained_score	INTEGER	DEFAULT 0	Obtained
maximum_score	INTEGER	DEFAULT 0	Maximum
state	JSONB	NULL	State payload

Chapter 8: Employer Domain

This chapter documents all Employer tables, columns, constraints and purposes.

Table: education.jobs

Column	Type	Constraint	Description
id	UUID	PK	Job id
employer_id	UUID	FK	References users
title	VARCHAR(255)	NOT NULL	Title
description	TEXT	NULL	Description
requirements	TEXT	NULL	Requirements
required_skills	TEXT[]	NULL	Skills
status	VARCHAR(50)	DEFAULT open	Status
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.applications

Column	Type	Constraint	Description
id	UUID	PK	Application id
job_id	UUID	FK	References jobs
student_id	UUID	FK	References users
status	VARCHAR(50)	DEFAULT submitted	Status
skill_match	DOUBLE PRECISION	NULL	Skill match
applied_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Applied

Chapter 9: Community Domain

This chapter documents all Community tables, columns, constraints and purposes.

Table: education.community_posts

Column	Type	Constraint	Description
id	UUID	PK	Post id
author_id	UUID	FK	References users
title	VARCHAR(255)	NOT NULL	Title
content	TEXT	NOT NULL	Content
post_type	ENUM	NOT NULL	Post type
status	ENUM	DEFAULT Published	Status
visibility	ENUM	DEFAULT Public	Visibility
media_url	TEXT	NULL	Media
metadata	JSONB	NULL	Metadata

comments_count	INTEGER	DEFAULT 0	Comments count
reactions_count	INTEGER	DEFAULT 0	Reactions count
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Updated
deleted_at	TIMESTAMP(6)	NULL	Soft delete

Table: education.community_comments

Column	Type	Constraint	Description
id	UUID	PK	Comment id
post_id	UUID	FK	References posts
author_id	UUID	FK	References users
parent_id	UUID	SELF FK NULL	Parent comment
content	TEXT	NOT NULL	Content
reactions_count	INTEGER	DEFAULT 0	Reactions
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Updated
deleted_at	TIMESTAMP(6)	NULL	Soft delete

Table: education.community_reactions

Column	Type	Constraint	Description
id	UUID	PK	Reaction id
user_id	UUID	FK	References users
post_id	UUID	FK NULL	References posts
comment_id	UUID	FK NULL	References comments
reaction_type	ENUM	DEFAULT Like	Reaction type
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.community_tags

Column	Type	Constraint	Description
id	UUID	PK	Tag id
name	VARCHAR(50)	UNIQUE	Tag name
slug	VARCHAR(50)	UNIQUE	Slug
usage_count	INTEGER	DEFAULT 0	Usage
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.community_post_tags

Column	Type	Constraint	Description
post_id	UUID	PK/FK	References posts
tag_id	UUID	PK/FK	References tags
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.community_bookmarks

Column	Type	Constraint	Description
id	UUID	PK	Bookmark id
user_id	UUID	FK	References users
post_id	UUID	FK	References posts
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Chapter 10: Administration Domain

This chapter documents all Administration tables, columns, constraints and purposes.

Table: education.notifications

Column	Type	Constraint	Description
id	UUID	PK	Notification id
user_id	UUID	FK	References users
type	VARCHAR(100)	NOT NULL	Type
message	TEXT	NOT NULL	Message
read_status	BOOLEAN	DEFAULT false	Read flag
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Table: education.subscriptions

Column	Type	Constraint	Description
id	UUID	PK	Subscription id
user_id	UUID	FK	References users
plan_type	VARCHAR(100)	NOT NULL	Plan
start_date	TIMESTAMP(6)	NOT NULL	Start
end_date	TIMESTAMP(6)	NOT NULL	End

Table: education.settings

Column	Type	Constraint	Description
id	UUID	PK	Setting id
scope	VARCHAR(100)	DEFAULT system	Scope
key	VARCHAR(255)	NOT NULL	Key
value	JSONB	NOT NULL	Value

Table: education.reports

Column	Type	Constraint	Description
id	UUID	PK	Report id
title	VARCHAR(255)	NOT NULL	Title
type	VARCHAR(100)	NOT NULL	Type
generated_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Generated
exported_at	TIMESTAMP(6)	NULL	Exported
format	VARCHAR(50)	NULL	Format
payload	JSONB	NULL	Payload

Table: education.announcements

Column	Type	Constraint	Description
id	UUID	PK	Announcement id
educator_id	UUID	FK	References users
title	VARCHAR(255)	NOT NULL	Title
message	TEXT	NOT NULL	Message
audience	VARCHAR(100)	DEFAULT all	Audience
scheduled_at	TIMESTAMP(6)	NULL	Schedule
attachment	TEXT	NULL	Attachment
created_at	TIMESTAMP(6)	DEFAULT CURRENT_TIMESTAMP	Created

Chapter 11: AI Support Domain

AI services do not own database storage. These backend-owned tables provide feature storage, course-skill mapping, prerequisites, learner ratings and recommendation support.

Table: education.activity_logs

Column	Type	Constraint	Description
id	UUID	PK	Activity id
user_id	UUID	FK	References users
sessions_last_30_days	INTEGER	NOT NULL	Sessions
avg_session_minutes	DOUBLE PRECISION	NULL	Avg session
videos_watched	INTEGER	NULL	Videos
assignments_attempted	INTEGER	NULL	Assignments
discussion_interactions	INTEGER	NULL	Discussions
login_count	INTEGER	NULL	Logins
device_count	INTEGER	NULL	Devices
ip_changes	INTEGER	NULL	IP changes
last_activity	TIMESTAMP	NULL	Last activity
suspicious_activity_score	NUMERIC	NULL	Suspicious score
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Updated

Table: education.login_history

Column	Type	Constraint	Description
id	UUID	PK	Login history id
user_id	UUID	FK	References users
logins_last_30_days	INTEGER	NULL	Login count
days_since_last_login	INTEGER	NULL	Inactivity
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Updated

Table: education.course_skills

Column	Type	Constraint	Description
id	UUID	PK	Mapping id
course_id	UUID	FK	References courses
skill_id	INTEGER	FK	References skills
skill_level	INTEGER	NULL	Level
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Updated

Table: education.course_prerequisites

Column	Type	Constraint	Description
id	UUID	PK	Mapping id
course_id	UUID	FK	References courses
prerequisite_course_id	UUID	FK	References courses
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Updated

Table: education.course_ratings

Column	Type	Constraint	Description
id	UUID	PK	Rating id
user_id	UUID	FK	References users
course_id	UUID	FK	References courses
rating	DOUBLE PRECISION	1-5	Rating
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Created
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Updated

Recommendation Table AI Enhancements

```
ALTER TABLE education.recommendations ADD COLUMN predicted_rating DOUBLE PRECISION;  
ALTER TABLE education.recommendations ADD COLUMN similarity_score DOUBLE PRECISION;  
ALTER TABLE education.recommendations ADD COLUMN confidence_score DOUBLE PRECISION;  
ALTER TABLE education.recommendations ADD COLUMN prerequisite_completed BOOLEAN;  
ALTER TABLE education.recommendations ADD COLUMN rank INTEGER;  
ALTER TABLE education.recommendations ADD COLUMN updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP;
```

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